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Digital Business and Technology Economics
Level: Intermediate to Advanced • Rev. 2026

SaaS Economics: MRR, ARR, Churn, and Net Revenue Retention (NRR)

Summary: Understand subscription software metrics. Learn how compounding Monthly Recurring Revenue, customer churn, and expansion revenue drive technology company valuation.

Learning Objectives

- Calculate Monthly Recurring Revenue (MRR) and Annual Recurring Revenue (ARR).
- Differentiate Gross Churn from Net Revenue Retention (NRR).
- Understand why negative net revenue churn creates exponential software enterprise value.
- Model subscription cohort decay and customer lifespan.

Core Principles & Key Takeaways

- MRR measures predictable, contracted monthly recurring subscription revenue.
- High customer churn acts as a leaky bucket that destroys acquisition ad spend.
- Net Revenue Retention (NRR) above 100% indicates strong product expansion and customer stickiness.

1. The Recurring Revenue Engine

Unlike traditional retail or custom service software firms that must resell to customers every single month, Software-as-a-Service (SaaS) charges a recurring subscription fee. This changes the economics: initial customer acquisition cost is high, but marginal delivery cost is near zero.

2. MRR Components and Formula

$$\text{Net New MRR} = \text{New MRR} + \text{Expansion MRR} - \text{Contraction MRR} - \text{Churned MRR}$$

$$\text{Annual Recurring Revenue (ARR)} = \text{MRR} \times 12$$

3. Net Revenue Retention (NRR) Formula

$$\text{NRR} = \frac{\text{Starting MRR} + \text{Expansion} - \text{Contraction} - \text{Churn}}{\text{Starting MRR}} \times 100\%$$

If $\text{NRR} > 100\%$, the software company grows its revenue from existing cohorts even if it acquires zero new customers! World-class enterprise SaaS companies achieve $\text{NRR} \ge 120\%$.

Self-Assessment & Review Questions

Q1: A SaaS startup starts January with ₦5,000,000 MRR from 100 clients. In January, existing clients upgrade by ₦800,000 (expansion), downgrades are ₦100,000, and cancellations are ₦200,000. What is their

NRR?

A. 90%
B. 110%
C. 100%
D. 125%

Correct Answer: B • NRR = (5,000,000 + 800,000 - 100,000 - 200,000) / 5,000,000 = 5,500,000 / 5,000,000 = 110%.

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